

A gold-grabbing game

OPG-37079

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Setup Fix a tree T and for every vertex $v \in V(T)$ a non-negative integer $g(v)$ which we think of as the amount of gold at v .

2-Player game Players alternate turns. On each turn, a player chooses a leaf vertex v of the tree, takes the gold at this vertex, and then deletes v . The game ends when the tree is empty, and the winner is the player who has accumulated the most gold.

Problem Find optimal strategies for the players.